

CHEMICAL BONDING & MOLECULAR STRUCTURE

EAPCET Practice Manual • 20 High-Yield Conceptual Problems with Solutions

This practice manual is designed to serve as a high-density, conceptual bridge between theoretical study and competitive exam practice. It contains exactly two carefully structured, multi-choice practice problems for each of the ten major concepts from the Chemical Bonding syllabus. Solving these problems will build the exact reasoning pathways, algebraic techniques, and logical frameworks required for students to confidently master the EAPCET PYQs in the main syllabus. Detailed, step-by-step conceptual explanations are provided at the end of the document.

PART I: PRACTICE PROBLEMS

Q1. Which of the following compounds is expected to exhibit the highest degree of covalent character according to Fajan's rules?

(A) NaCl

(B) CuCl

(C) KCl

(D) RbCl

Q2. Arrange the following ionic solids in the correct order of increasing lattice enthalpy (smallest to largest): NaCl, MgO, CaCl_2 , NaF.

(A) $\text{NaCl} < \text{NaF} < \text{CaCl}_2 < \text{MgO}$

(B) $\text{NaF} < \text{NaCl} < \text{CaCl}_2 < \text{MgO}$

(C) $\text{NaCl} < \text{CaCl}_2 < \text{NaF} < \text{MgO}$

(D) $\text{MgO} < \text{CaCl}_2 < \text{NaF} < \text{NaCl}$

Q3. Which of the following molecular species contains a central atom that does not obey the octet rule, specifically by possessing an incomplete octet (electron-deficient central atom)?

(A) CH_4

(B) BF_3

(C) NH_3

(D) PCl_5

Q4. Using the formal charge expression $Q = [V] - [L] - \frac{1}{2}[B]$, calculate the formal charge on the central oxygen atom in the resonance structure of the ozone (O_3) molecule.

(A) -1

(B) 0

(C) +1

(D) +2

Q5. According to VSEPR theory, what is the molecular shape of XeF_4 and the total number of lone pairs residing on the central Xenon atom?

(A) Octahedral, 0 lone pairs

(B) Square planar, 2 lone pairs

(C) Tetrahedral, 0 lone pairs

(D) Square pyramidal, 1 lone pair

Q6. The H-O-H bond angle in water (H_2O) is 104.5° , whereas the H-S-H bond angle in hydrogen sulfide (H_2S) is 92° . Which of the following statements correctly explains this difference?

(A) Oxygen is less electronegative than sulfur, which increases repulsion between hydrogen atoms.

(B) The lone pairs on sulfur are highly localized, pushing the S-H bond pairs further apart.

(C) The lower electronegativity and larger size of the central sulfur atom shift the bond pairs further away, reducing the bp-bp repulsion and allowing lone pairs to compress the bond angle more strongly.

(D) Intermolecular hydrogen bonding in H_2S forces the molecules to adopt a tighter bond angle.

Q7. Which of the following compounds has a central atom in the exact same hybridisation state as the central phosphorus atom in PCl_5 ?

(A) SF_6

(B) XeF_4

(C) SF_4

(D) BF_3

Q8. Arrange the carbon atoms in ethyne (C_2H_2), ethene (C_2H_4), and ethane (C_2H_6) in order of increasing electronegativity.

(A) ethane < ethene < ethyne

(B) ethyne < ethene < ethane

(C) ethene < ethane < ethyne

(D) All carbon atoms possess identical electronegativity regardless of hybridisation.

Q9. While carbon dioxide (CO_2) has a net dipole moment of zero, sulfur dioxide (SO_2) has a significant permanent dipole moment ($\mu = 1.62 \text{ D}$). This structural divergence is because:

(A) CO_2 contains non-polar bonds, whereas SO_2 contains polar bonds.

(B) CO_2 is linear and its bond dipoles cancel, whereas SO_2 is bent due to a central lone pair, preventing dipole cancellation.

(C) Sulfur is much more electronegative than carbon, leading to a much stronger net pull in SO_2 .

(D) CO_2 is a planar cyclic molecule, whereas SO_2 is tetrahedral.

Q10. The experimental dipole moment of a diatomic polar molecule HX is observed to be 1.20 D . If the bond length is measured to be $1.00 \text{ \AA}$, estimate the percentage ionic character of the HX bond. (Given: electronic charge $e = 4.80 \times 10^{-10} \text{ esu}$, $1 \text{ \AA} = 10^{-8} \text{ cm}$, and $1 \text{ D} = 10^{-18} \text{ esu}\cdot\text{cm}$).

(A) 15%

(B) 25%

(C) 50%

(D) 75%

Q11. Consider the carbon-oxygen bonds in carbon monoxide (CO), carbon dioxide (CO_2), and the carbonate ion (CO_3^{2-}). What is the correct order of increasing carbon-oxygen bond length?

(A) $\text{CO} < \text{CO}_2 < \text{CO}_3^{2-}$

(B) $\text{CO}_3^{2-} < \text{CO}_2 < \text{CO}$

(C) $\text{CO}_2 < \text{CO} < \text{CO}_3^{2-}$

(D) $\text{CO} < \text{CO}_3^{2-} < \text{CO}_2$

Q12. Which of the following represents a standard general property of covalent compounds?

(A) They possess exceptionally high melting and boiling points due to strong lattice forces.

(B) They are excellent electrical conductors in both solid and molten states.

(C) They are generally characterized by relatively low melting and boiling points, and are poor electrical conductors.

(D) All covalent compounds, without exception, dissolve readily in polar solvents like water.

Q13. According to Molecular Orbital Theory (MOT), which of the following diatomic oxygen species possesses the shortest bond length and is paramagnetic?

(A) O_2^{2-}

(B) O_2^-

(C) O_2

(D) O_2^+

Q14. According to MOT, which of the following homonuclear diatomic species cannot exist in stable form under standard conditions?

(A) Li_2

(B) B_2

(C) Be_2

(D) C_2

Q15. Compare the chemical structures of ortho-nitrophenol and para-nitrophenol. Which compound is expected to have the higher boiling point, and what type of hydrogen bonding is responsible?

(A) ortho-Nitrophenol, due to intramolecular hydrogen bonding

(B) para-Nitrophenol, due to intermolecular hydrogen bonding

(C) ortho-Nitrophenol, due to intermolecular hydrogen bonding

(D) para-Nitrophenol, due to intramolecular hydrogen bonding

Q16. The hydrides of Group 15 (NH_3), Group 16 (H_2O), and Group 17 (HF) exhibit unusually high boiling points compared to the heavier hydrides in their respective groups. This anomaly is due to:

(A) Strong covalent bonds formed between separate gaseous molecules.

(B) High molecular mass leading to massive dispersion forces.

(C) Extensive intermolecular hydrogen bonding requiring substantial energy to break.

(D) Their non-polar nature which leads to stable crystal packing.

Q17. Which of the following statements regarding the bonding and spatial structure of diborane (B_2H_6) is correct?

(A) It contains six equivalent terminal B-H bonds lying in a single horizontal plane.

(B) It consists of two terminal B-H-B banana bonds and four bridging covalent bonds.

(C) It contains four terminal two-centre two-electron (2c-2e) B-H bonds and two bridging three-centre two-electron (3c-2e) B-H-B banana bonds.

(D) Boron atoms are directly bonded to each other by a strong sigma bond.

Q18. Classify the hydrides B_2H_6 , CH_4 , and NH_3 according to their bonding characteristics and electron availability.

(A) B_2H_6 is electron-precise, CH_4 is electron-deficient, NH_3 is electron-rich.

(B) B_2H_6 is electron-deficient, CH_4 is electron-precise, NH_3 is electron-rich.

(C) B_2H_6 is electron-rich, CH_4 is electron-precise, NH_3 is electron-deficient.

(D) All three are electron-precise hydrides that strictly obey the Lewis octet rule.

Q19. Which of the following sets represents a group of species that are entirely isoelectronic with one another?

(A) N_2 , CO , CN^- , NO^+

(B) O_2 , CO , N_2 , HF

(C) CO_2 , SO_2 , NO_2 , O_3

(D) Na^+ , Mg^{2+} , Cl^- , Ar

Q20. Isoelectronic species like N_2 and CO share remarkably similar physical and structural properties. The fundamental reason for this similarity is:

(A) They possess identical molecular masses, leading to equal kinetic energy distributions.

(B) They have the exact same electronegativity profile and atomic sizes.

(C) They possess the same total number of electrons, leading to identical molecular orbital configurations, the same bond order of 3, and a triple-bond structure.

(D) They are made of elements from the same period of the periodic table.

PART II: DETAILED SOLUTIONS

Solution to Q1: Correct Option (B)

According to **Fajan's rules**, the covalent character of an ionic bond is favored by having a highly polarizing cation and a highly polarizable anion. Cation polarizing power increases with smaller size and higher charge. In this set, Cu^+ has a pseudo-noble-gas configuration ($[\text{Ar}] 3d^{10}$, 18 electrons in the outermost shell), which is much less effective at shielding nuclear charge than an inert-gas configuration (like Na^+ , $[\text{Ne}] 2s^2 2p^6$, 8 electrons). Consequently, Cu^+ possesses a higher effective nuclear charge and greater polarizing power than Na^+ of similar size. This induces a much higher degree of covalent character in **CuCl** than in NaCl, KCl, or RbCl, which have inert-gas configurations.

Solution to Q2: Correct Option (A)

Lattice enthalpy ($\Delta H_{\text{lattice}}$) is directly proportional to the product of ionic charges ($|z^+z^-|$) and inversely proportional to the interionic distance ($r^+ + r^-$). Let us evaluate the parameters for each solid:

- **MgO**: $z^+ = +2$, $z^- = -2$. Charge product = 4. Due to the double charge and relatively small size of Mg^{2+} and O^{2-} , it has the highest lattice enthalpy.
- **CaCl₂**: $z^+ = +2$, $z^- = -1$. Charge product = 2. It has higher lattice enthalpy than 1:1 electrolytes.
- **NaF** and **NaCl**: Both are 1:1 electrolytes (charge product = 1). Since F^- is smaller than Cl^- , the interionic distance is shorter in NaF, leading to stronger electrostatic attractions and higher lattice enthalpy than NaCl. Thus, the correct increasing order of lattice enthalpy is: **NaCl < NaF < CaCl₂ < MgO**.

Solution to Q3: Correct Option (B)

The **octet rule** states that atoms tend to form bonds to achieve 8 electrons in their valence shell. • **BF₃**: Boron is in Group 13 and has 3 valence electrons. When it forms 3 single covalent bonds with F atoms, it shares 6 electrons in total. There are no lone pairs on the central B atom, leaving it with 6 valence electrons—an **incomplete octet** (electron-deficient).

- **CH₄** and **NH₃**: Carbon and nitrogen successfully complete their octets (8 valence electrons).
- **PCl₅**: Phosphorus undergoes an **expanded octet** containing 10 valence electrons in its bonded state, which violates the octet rule but via hypervalency rather than deficiency.

Solution to Q4: Correct Option (C)

The formal charge (Q) is calculated as: **$Q = [\text{V}] - [\text{L}] - \frac{1}{2}[\text{B}]$** .

In the ozone (O_3) molecule, the central oxygen atom ($\text{O}_{\text{central}}$) is connected to one terminal oxygen by a double bond and to the other by a single bond, while carrying a lone pair:

- Valence electrons of free oxygen atom, **$[\text{V}] = 6$** .
- Non-bonding (lone pair) electrons on the central atom, **$[\text{L}] = 2$** (1 lone pair).
- Bonding (shared) electrons, **$[\text{B}] = 6$** (3 bonding pairs total: 1 double bond + 1 single bond).

Applying the formula: $Q = 6 - 2 - \frac{1}{2}(6) = 6 - 2 - 3 = +1$.

This positive charge on the central oxygen is balanced by a -1 charge on the single-bonded terminal oxygen, yielding a neutral net molecular charge.

Solution to Q5: Correct Option (B)

To determine the shape of XeF_4 , we use VSEPR theory:

- Xenon is a noble gas with **8 valence electrons**. It forms 4 covalent bonds with 4 fluorine atoms, using 4 electrons.
- The remaining 4 electrons form **2 lone pairs** on the central Xe atom.
- Steric number = 4 bond pairs (BP) + 2 lone pairs (LP) = 6. This corresponds to an octahedral electron-pair geometry.
- To minimize the strong lone-pair-lone-pair (lp-lp) repulsions, the 2 lone pairs occupy the axial, trans positions (180° apart).
- The 4 fluorine atoms occupy the equatorial positions, resulting in a symmetrical **square planar** molecular geometry with a net dipole moment of zero.

Solution to Q6: Correct Option (C)

Both H_2O and H_2S have central atoms with 2 bond pairs and 2 lone pairs (steric number = 4, tetrahedral arrangement). However, the bond angle in H_2S (92°) is much smaller than in H_2O (104.5°):

- Oxygen is highly electronegative and small; it draws the O-H bonding electron pairs close to itself. The high density of bonding electrons near the oxygen atom causes significant **bond pair-bond pair (bp-bp) repulsion**, which resists further compression by the lone pairs, holding the angle at 104.5° .
- Sulfur is larger and much less electronegative than oxygen. The S-H bonding electron pairs lie further away from the sulfur nucleus, towards the hydrogen atoms. This vastly reduces the bp-bp electrostatic repulsion near the central atom, allowing the strong lone pair-lone pair (lp-lp) and lone pair-bond pair (lp-bp) repulsions to compress the H-S-H bond angle tightly down to nearly 90° (specifically, 92°).

Solution to Q7: Correct Option (C)

Let us determine the hybridisation of the central atoms:

- PCl_5 : Central phosphorus (P) has 5 valence electrons and forms 5 bond pairs (0 lone pairs). Steric number = 5, which corresponds to **sp^3d** hybridisation.
- SF_4 : Central sulfur (S) has 6 valence electrons. It forms 4 bond pairs and has 1 lone pair. Steric number = $4 + 1 = 5$, which corresponds to **sp^3d** hybridisation. Thus, SF_4 has the exact same hybridisation state as PCl_5 (though its molecular shape is see-saw instead of trigonal bipyramidal).
- SF_6 and XeF_4 are sp^3d^2 hybridised (steric number = 6).
- BF_3 is sp^2 hybridised (steric number = 3).

Solution to Q8: Correct Option (A)

The electronegativity of an atom is directly proportional to the **s-character** of its hybrid orbitals, because s-orbitals are closer to the nucleus and experience a stronger effective nuclear charge. Let us evaluate the hybridisation states of carbon in these molecules:

- **ethane (C_2H_6)**: Carbon is single-bonded to four atoms, corresponding to **sp^3** hybridisation (25% s-character).
- **ethene (C_2H_4)**: Carbon is double-bonded, corresponding to **sp^2** hybridisation (33.3% s-character).
- **ethyne (C_2H_2)**: Carbon is triple-bonded, corresponding to **sp** hybridisation (50% s-character).

As s-character increases from 25% to 50%, the electronegativity of carbon increases significantly. Therefore, the correct order is: **ethane (sp^3) < ethene (sp^2) < ethyne (sp)**.

Solution to Q9: Correct Option (B)

CO_2 and SO_2 are both triatomic molecules with polar bonds, but they differ fundamentally in geometry:

- CO_2 has a central carbon with 2 double bonds and 0 lone pairs (sp hybridised). The molecule is completely linear ($\text{O}=\text{C}=\text{O}$). The two C-O bond dipoles are equal in magnitude but point in exactly opposite directions (180°), completely canceling each other out and yielding a **net dipole moment of zero** ($\mu = 0$).
- SO_2 has a central sulfur with 2 double bonds and 1 lone pair (sp^2 hybridised). This lone pair exerts repulsion, forcing the S-O bonds down into a **bent (V-shaped) geometry**. Because the bond dipoles point at an angle (roughly 120°), they reinforce each other rather than canceling, resulting in a **permanent net dipole moment** ($\mu = 1.62 \text{ D}$).

Solution to Q10: Correct Option (B)

To find the percentage ionic character of the bond in HX, we compare the observed dipole moment (μ_{observed}) to the theoretical dipole moment assuming 100% ionic transfer ($\mu_{\text{theoretical}}$):

1. Calculate $\mu_{\text{theoretical}} = e \times d$, where the charge transferred is one full electronic charge:

$$e = 4.80 \times 10^{-10} \text{ esu}$$

$$d = 1.00 \text{ \AA} = 1.00 \times 10^{-8} \text{ cm}$$

$$\mu_{\text{theoretical}} = (4.80 \times 10^{-10} \text{ esu}) \times (1.00 \times 10^{-8} \text{ cm}) = 4.80 \times 10^{-18} \text{ esu}\cdot\text{cm} = \mathbf{4.80 \text{ D}}$$
 (since $1 \text{ D} = 10^{-18} \text{ esu}\cdot\text{cm}$).

2. Calculate percentage ionic character:

$$\% \text{ Ionic Character} = [\mu_{\text{observed}} / \mu_{\text{theoretical}}] \times 100\%$$

$$\% \text{ Ionic Character} = [1.20 \text{ D} / 4.80 \text{ D}] \times 100\% = 0.25 \times 100\% = \mathbf{25\%}.$$

Solution to Q11: Correct Option (A)

Bond length is inversely proportional to **Bond Order** (the number of chemical bonds between a pair of atoms). Let us determine the carbon-oxygen bond orders in the three species:

- **CO (Carbon Monoxide)**: Exists with a triple bond ($\text{C}\equiv\text{O}$). **Bond Order = 3.0**.
- **CO_2 (Carbon Dioxide)**: Exists as a linear molecule with two double bonds ($\text{O}=\text{C}=\text{O}$). **Bond Order = 2.0**.
- **CO_3^{2-} (Carbonate Ion)**: Resonance occurs across three equivalent carbon-oxygen positions. The total of 4 bonding pairs is shared equally among 3 locations, yielding an average **Bond Order = $4/3 \approx 1.33$** .

Since bond length increases as bond order decreases, the correct order of increasing carbon-oxygen bond length is: **CO (shortest) < CO_2 < CO_3^{2-} (longest)**.

Solution to Q12: Correct Option (C)

Covalent compounds consist of discrete molecules held together by relatively weak intermolecular forces (such as London dispersion forces or dipole-dipole attractions). Because these weak forces are easily overcome with thermal energy, typical covalent compounds possess **relatively low melting and boiling points**. Furthermore, because they do not contain free, mobile ions or delocalized electrons in their solid or liquid states, they are **poor electrical conductors**. Note that high solubility in polar solvents like water is not a general property of covalent compounds; many are highly hydrophobic and insoluble (e.g., hydrocarbons, lipids, sulfur) unless they can form strong hydrogen bonds.

Solution to Q13: Correct Option (D)

According to MOT, we can determine the properties of the diatomic oxygen species by counting their electrons and filling the molecular orbitals:

- O_2 (16 e^-): Configuration has antibonding π^*_{2p} (two unpaired electrons in degenerate π^* orbitals). **Bond Order** = $\frac{1}{2}(10 - 6) = 2.0$. It is paramagnetic.
- O_2^{2-} (18 e^-): Added 2 electrons to antibonding π^* orbitals (π^*_{2p} , all paired). **Bond Order** = $\frac{1}{2}(10 - 8) = 1.0$. It is diamagnetic.
- O_2^- (17 e^-): Added 1 electron to antibonding π^* orbitals (π^*_{2p} , 1 unpaired). **Bond Order** = $\frac{1}{2}(10 - 7) = 1.5$. It is paramagnetic.
- O_2^+ (15 e^-): Removed 1 electron from an antibonding π^* orbital (π^*_{2p} , 1 unpaired). **Bond Order** = $\frac{1}{2}(10 - 5) = 2.5$. It is paramagnetic.

Since bond length is inversely related to bond order, the species with the highest bond order (O_2^+ , **BO** = 2.5) has the shortest bond length. It is also paramagnetic (1 unpaired electron).

Solution to Q14: Correct Option (C)

For a homonuclear diatomic molecule to exist in stable form, its **Bond Order must be greater than zero**. Let us calculate the bond order for Be_2 :

- Beryllium has atomic number $Z = 4$. Thus, Be_2 would contain **8 electrons**.
- Filling the molecular orbitals in order of increasing energy: $\sigma_{1s}^2, \sigma_{1s}^{*2}, \sigma_{2s}^2, \sigma_{2s}^{*2}$.
- Number of bonding electrons, $N_b = 4$ (from σ_{1s} and σ_{2s}).
- Number of antibonding electrons, $N_a = 4$ (from σ_{1s}^* and σ_{2s}^*).
- Bond Order = $\frac{1}{2}(N_b - N_a) = \frac{1}{2}(4 - 4) = 0.0$.

A bond order of zero indicates that there is no net stabilization energy to hold the two beryllium atoms together, meaning the diatomic Be_2 molecule cannot exist under standard conditions.

Solution to Q15: Correct Option (B)

The difference in boiling points between these isomers lies in the spatial orientation of their groups:

- **ortho-Nitrophenol**: The hydroxyl (-OH) and nitro ($-\text{NO}_2$) groups are on adjacent carbon atoms. This close proximity allows the hydrogen of the -OH group to form an **intramolecular hydrogen bond** with the oxygen of the $-\text{NO}_2$ group, creating a stable 6-membered chelate ring. Since the hydrogen bonding is locked within the molecule, these molecules cannot associate strongly with neighboring molecules. This makes ortho-nitrophenol steam-volatile with a lower boiling point.
- **para-Nitrophenol**: The -OH and $-\text{NO}_2$ groups are on opposite sides of the benzene ring. Intramolecular bonding is geometrically impossible. Instead, the -OH group of one molecule forms strong **intermolecular hydrogen bonds** with the $-\text{NO}_2$ groups of separate neighboring molecules, creating an extensive associated molecular network. Overcoming these strong intermolecular forces requires substantial thermal energy, giving para-nitrophenol a **significantly higher boiling point**.

Solution to Q16: Correct Option (C)

Under normal trends, boiling points of group hydrides increase down the group because larger molecules have larger electron clouds, creating stronger London dispersion (Van der Waals) forces (e.g., $\text{H}_2\text{S} < \text{H}_2\text{Se} < \text{H}_2\text{Te}$). However, NH_3 , H_2O , and HF deviate dramatically, showing extremely high boiling points.

This is because N, O, and F are exceptionally small and highly electronegative atoms. When covalently bonded to hydrogen, they create highly polar bonds, and their lone pairs form extensive, exceptionally strong networks of **intermolecular hydrogen bonding**. These highly associated molecular structures require massive quantities of kinetic energy to break apart into the gas phase, raising their boiling points far above those of the heavier hydrides in their respective groups.

Solution to Q17: Correct Option (C)

Diborane (B_2H_6) is an electron-deficient molecule with a unique bridged structure:

- Each boron atom undergoes sp^3 hybridisation.
- There are **four terminal hydrogen atoms**. Each terminal H is bonded to a boron atom by a standard, localized, two-centre two-electron (2c-2e) covalent bond. These four terminal B-H bonds and the two B atoms all lie in a single plane.
- There are **two bridging hydrogen atoms**. The bridge is formed by two **three-centre two-electron (3c-2e) B-H-B bonds**, which are commonly referred to as **banana bonds**. In these bonds, a pair of bonding electrons is shared across three nuclei (two Borons and one Hydrogen).
- These bridging bonds lie in a plane perpendicular to the terminal atoms (one above and one below), stabilizing the electron-deficient boron atoms without any direct B-B bond.

Solution to Q18: Correct Option (B)

Hydrides can be classified into three groups based on their electron and orbital counts:

- **Electron-Deficient (e.g., B_2H_6)**: Hydrides that do not possess a sufficient number of valence electrons to form standard two-electron covalent bonds between all adjacent atoms. Group 13 hydrides are electron-deficient and must undergo complex bridging (like diborane) to stabilize.
- **Electron-Precise (e.g., CH_4)**: Hydrides that have exactly the correct number of valence electrons to form traditional localized single bonds with all surrounding hydrogen atoms, with no leftover lone pairs. Group 14 hydrides are electron-precise.
- **Electron-Rich (e.g., NH_3)**: Hydrides that have more valence electrons than required for bonding, resulting in one or more lone pairs on the central atom (Group 15, 16, and 17 hydrides).

Solution to Q19: Correct Option (A)

Isoelectronic species are atoms, molecules, or ions that contain the **exact same total number of electrons**. Let us count the total electrons for the species in set A:

- N_2 : Nitrogen has atomic number 7. Total electrons = $7 + 7 = 14$ **electrons**.
- CO : Carbon (6) and Oxygen (8). Total electrons = $6 + 8 = 14$ **electrons**.
- CN^- : Carbon (6), Nitrogen (7), plus 1 negative charge. Total electrons = $6 + 7 + 1 = 14$ **electrons**.
- NO^+ : Nitrogen (7), Oxygen (8), minus 1 positive charge. Total electrons = $7 + 8 - 1 = 14$ **electrons**.

Since all four species have exactly 14 electrons, they form a complete isoelectronic set. (As a result, they also share a triple-bond configuration, a bond order of 3, and similar molecular shapes).

Solution to Q20: Correct Option (C)

Because isoelectronic species like N_2 and CO have the exact same number of electrons (14 electrons total, with 10 valence electrons), they share **identical molecular orbital diagrams** and electronic configurations. Both fill their molecular orbitals to yield a stable triple bond consisting of one σ bond and two π bonds, giving them an identical **Bond Order of 3.0**. Since their outermost electron distributions and bonding strengths are nearly identical, they share extremely similar physical properties (such as exceptionally low boiling points and high chemical stabilities) and identical linear geometries.